Measurement and simulation of Transmon-type superconducting qubits in a 3D cavity: characterization of device energies and characteristic times T_1 and T_2^*

Final Report

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Abstract

In this research internship project, we continue the steps from previous internships, where the theory and experience in simulating quantum systems are applied through real measurements and characterization of superconducting junction transmon quantum bits. This includes obtaining their parameters, such as the resonance frequencies of the qubit and cavity, ω_{01}, $\omega_{02}/2$, and ω_C, the coupling parameters χ and g, as well as the relaxation and dephasing times T_1 and T_2^*. These will then be validated through simulations using QuTiP. The work also includes auxiliary tasks such as improving the code executed for the measurement procedures and helping with testing, characterizing, and optimizing other experimental instruments. It also includes several simulations of the effects of drive noise in the measurement procedures of a qubit, which were additional work beyond the internship proposal. This work on noise simulation is being developed into a publication by the intern and some of his laboratory colleagues.

Introduction

A GitHub repository with the project files and notebooks is available here. Given the page constraints of this report, it is recommended to visit the repository for more information on the work conducted during this internship, with emphasis on the simulation notebooks. Links to specific sections are occasionally provided throughout the report.

1 Superconducting qubits

The kind of device used in our laboratory to produce the approximate quantum two-level system that can be used as a qubit is a superconducting Josephson junction non-harmonic oscillator, specifically, the transmon.

These devices work similarly to a quantum harmonic oscillator. A quantum harmonic oscillator, unlike a qubit, has infinitely many discrete energy levels, each separated by the same amount of energy. We can understand this quantum harmonic oscillator as the quantized equivalent of a classical harmonic oscillator, such as an inductor connected to a capacitor in parallel (Figure 1), where the energy oscillates back and forth between the charge of the capacitor and the magnetic flux of the inductor. In the quantum case, the energy is limited to discrete levels-modes of a confined wave (Figure 2). The wavefunction, in this case, is confined to the parabolic potential energy well of the harmonic oscillator.

The difference between this harmonic oscillator and the qubit is that the qubit has a Josephson junction in place of the inductor. This junction functions like an inductor but introduces a non-linearity that causes the energy levels of the quantum system to no longer be equally spaced. Much like the energy levels of an atom, this gives rise to

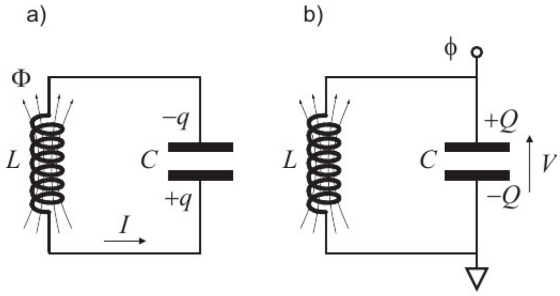

Figure 1: Adapted from [1]. a) State of the oscillator with the energy stored in the inductor's magnetic flux and maximum current. b) Energy stored in the capacitor's charge and maximum voltage. In a mass-spring harmonic oscillator, the energy in the capacitor would be analogous to potential energy, and (b) would correspond to the maximum stretching of the spring; the energy in the inductor would be analogous to kinetic energy, and (a) would correspond to maximum speed. The voltage would also correspond to position, and the current, to velocity.

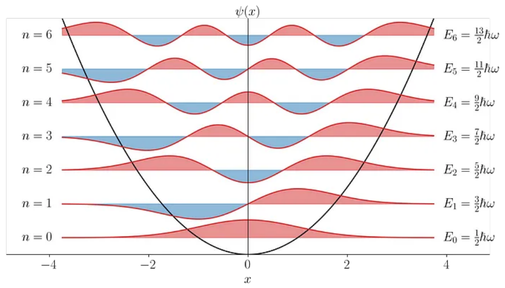

Figure 2: Adapted from [2]. Wavefunctions of the energy levels of the quantum harmonic oscillator, superposed on the quadratic potential well. For each energy level, we list its energy and its quantum number n.

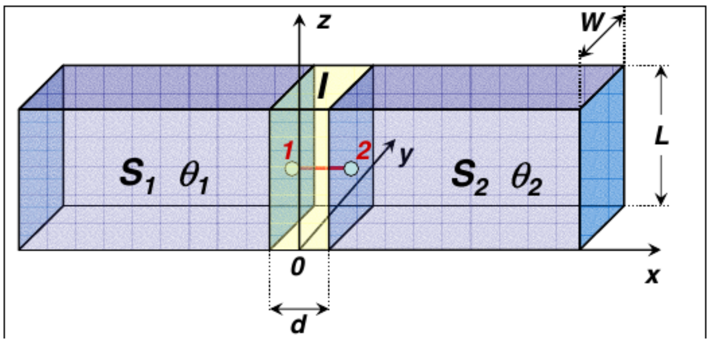

Figure 3: Adapted from [3]. Schematic of a Josephson junction. The blue blocks represent superconductors separated by an insulating gap. The junction is driven by a current, and each block has a distinct phase in the wavefunction of the Cooper pairs. The phase difference between the two blocks has a direct relationship to the current flowing through the junction, and the rate of change of this phase is determined by the voltage across the junction.

the term artificial atom for such a system. This energy anharmonicity also allows us to excite the first energy level independently of the others, enabling us to treat the system as a two-level one under certain conditions. See [1].

The Josephson junction (Figure 3) is a superconducting wire with a tiny gap of insulating material separating the two regions of the superconductor. The supercurrent of Cooper pairs of electrons is able to tunnel through this gap, resulting in the behavior of the non-linear inductor mentioned earlier [3].

To drive our qubit, we can directly apply a signal to the system, which induces oscillations between the quantum levels when the signal frequency is close to the qubit's transition frequency—much like a light-matter interaction where atoms are understood as a two-level system. This frequency is defined by the energy of the first-level transition. To measure our system, we couple the qubit with a metal cavity, which behaves like a quantum harmonic oscillator when alone (bare cavity) and confines an electromagnetic field, having a defined transition frequency (energy) between its discrete energy states. When coupled with the qubit, the entire system can be described by the Jaynes-Cummings Hamiltonian approximation. In the proper regime (the dressed cavity), this model shows how changing the qubit's state from ground to excited also changes the cavity's harmonic transition frequency. Thus, by probing the cavity at its transition frequency—where the magnitude of the signal coming out of it changes—we can infer the measured state of the qubit by checking whether the probe signal output corresponds to that of a cavity at its transition frequency [4].

The theory of superconducting qubits with Josephson junctions is explored in greater depth in the final report of the previous internship conducted in this laboratory and is available on this GitHub repository.

These qubits are interesting for applications such as quantum computing as long as their quality parameters are sufficiently good. This ensures they remain operational as quantum two-level systems long enough to perform all intended operations, such as several quantum logic gates. The loss of the quantum mechanical nature of our system is due to effects such as decoherence and other forms of external interference. The main parameters used to determine the quality of our samples are the relaxation time (T_1), which is the time constant of the natural exponential decay of the system from the excited state to the ground state, and the dephasing time (T_2), which measures the exponential time constant for the loss of the qubit's phase coherence, causing it to settle from a superposition into a defined state.

Methodology and activities

2 The measurement procedures

The equipment setup and procedures used to measure the quantum properties of superconducting transmon qubits are similar to those used in [4].

2.1 The VNA setup

In this setup, we use a VNA (Vector Network Analyzer), which scans the cavity's input with multiple frequencies and collects the phase and amplitude of the cavity's output. To excite the qubit, we connect the drive wire terminal directly to an RF source controlled by our computer.

2.2 Finding the cavity

The first step in measuring a sample is to find the cavity's excitation frequency. We observe the spectrum around the expected frequency, looking for a dip where the cavity absorbs the signal, and also check for the characteristic phase transition caused by the cavity's absorption at this dip to confirm it is the correct frequency. The next step to ensure we are at the cavity's transition frequency is the punchout procedure.

2.3 The punchout

With the punchout, we scan the cavity frequency again around the transition value using the VNA. But this time, we vary the amplitude of our input signal by using a variable attenuator mounted before the cavity input. At low signal intensity, the cavity is in the dressed state and has a certain transition frequency because it interacts with the qubit. However, if we increase the intensity beyond a certain threshold, we overwhelm the qubit with photons, inducing a current much higher than the critical current of the junction. This effectively changes the cavity to the bare state (not coupled to the qubit), thus changing its transition frequency. Therefore, if there is a working qubit on our sample, we observe distinct cavity transition frequencies at high and low power [4].

2.4 The Two Tone Spectroscopy

Now that we know the cavity's transition frequency and a cavity signal intensity at the desired regime (which corresponds to low attenuation), we can try to find the qubit's experimental transition frequency. To do this, we continue using the VNA but collect data points only at the cavity's transition. Effectively, if the cavity's transition changes because the qubit's state goes to excited, the cavity stops absorbing the signal within the small frequency range defined by the VNA at the peak of the cavity transition, and the output signal increases. Therefore, by exciting the qubit with a signal and scanning the frequency of this signal, we will observe a peak at the cavity output when the qubit signal frequency drives it to the excited state. By scanning the qubit frequency around the expected value inferred from the fabrication procedure, we can determine the experimental qubit transition frequency.

We can also observe half the frequency of the transition from the ground state to the second energy level of the qubit by using a high enough intensity of the qubit signal, although with much lower contrast. This occurs because a two-photon transition can happen between these energy levels. Using this transition, we can determine the anharmonicity of our qubit—that is, how much it deviates from a regular harmonic oscillator.

2.5 The pulsed setup

To continue the characterization procedure of the qubit, we now switch to the pulsed setup. Instead of using the VNA to measure the cavity response, we generate pulses using a switch capable of producing arbitrary signals with very fine resolution. To collect the data, we use an Alazar capture board that can sample at rates on the order of hundreds of MHz.

The signal sent to the cavity is processed using the heterodyne technique to reduce the frequency of its output signal. This allows us to capture the phase and amplitude of the signal with our capture board at a low sampling rate, while still reaching the excitation frequency of the cavity. We do this by generating a signal on an RF source around the cavity frequency (approximately 7 GHz) and mixing it with a 240 MHz signal. The mixed signal—consisting of the RF source frequency plus 240 MHz—is then sent to the cavity, where its phase and amplitude are modified according to the quantum system's parameters. The cavity's output signal is then mixed again with the original RF source signal. By filtering the difference between the two mixer inputs, we recover a 240 MHz signal carrying the amplitude and phase information imprinted by the cavity. This signal is captured by the capture board and digitally processed using the homodyne technique to extract the quadratures I and Q, which provide information about the phase and amplitude of our signal.

The signal for the qubit can be generated in several ways. The two methods used during my internship were: first, generating DC square pulses on the I and Q ports of an IQ mixer using a switch, with the IF input provided by an RF source. This setup allows us to control the amplitude and phase of the RF source signal and shape it into pulses. However, a problem with this technique was that when the I and Q voltages were zero, some signal still passed through the mixer, interfering with the qubit during periods such as measurement. We called this the "leakage." The other method was generating the RF signal pulses for the qubit directly from the switch. However, for some reason, this setup caused the dilution refrigerator stage where the sample was located to start heating.

We then repeat the punchout and two-tone procedures using the pulsed setup to verify that everything is functioning properly.

2.6 The Rabi Oscillation

With the experimental excitation frequencies of the qubit and the cavity determined, we can start performing Rabi oscillations. To do this, we set up a pulse sequence where we apply an excitation pulse to the qubit at its frequency with a variable duration, followed by a measurement pulse. By averaging several pulse sequences, we obtain the expected value of the qubit's state for each pulse duration. The behavior of this expected value as a function of the pulse duration follows a damped cosine, characteristic of the Rabi oscillation of a quantum system.

By fitting the data to a damped cosine or simply measuring the time duration of the first peak, we can determine the period of the Rabi oscillation. Half of this period corresponds to the π pulse duration, which excites the qubit from the ground state to the excited state. A $\pi/2$ pulse, or a quarter of the period, changes the qubit from the ground state to a superposition state.

The Rabi oscillation also depends on the detuning Δ from the resonance frequency of the qubit and the amplitude of the drive signal A [4]:

$$P_e(t) = \frac{A^2}{A^2+\Delta^2}\sin^2\left(\frac{\sqrt{A^2+\Delta^2}}{2}t\right)$$

This equation describes an isolated system and therefore does not include the damping observed in the experimental oscillation.

2.7 The T_1 measurement

To perform this measurement, we use a π pulse to excite the qubit and then wait for a certain duration before measuring the qubit state. The expected value of this measurement exhibits an exponential decay as a function of the time interval between excitation and measurement. The time constant of this decay is our T_1 parameter.

2.8 The T_2^* measurement or Ramsey Interferometry

To perform Ramsey interferometry, which is also used in atomic clocks, we apply two $\pi/2$ pulses separated by a duration T. The expected value of the qubit state varies according to a damped cosine oscillation as a function of this period T. The frequency of this oscillation depends on the detuning of the pulses from the qubit's resonance frequency. If the excitation signal is perfectly on resonance, the oscillation frequency is zero; therefore, we need to excite the qubit with a slight detuning to perform this measurement.

From the damped cosine's decay time constant we get our T_2^* parameter which is less or equal to the actual T_2.

2.9 The $T_2^{\mathbf{ECHO}}$ measurement

In this measurement, we perform a procedure similar to Ramsey interferometry but include a refocusing π pulse exactly in the middle of the two $\pi/2$ pulses, with two equal intervals of length T between the pulses. This refocusing pulse reverses the direction of the qubit's precession when it is in superposition, effectively canceling out the precession before the π pulse with the precession after it. As a result, the expected value of the measurement shows only an exponential decay rather than the oscillation observed in Ramsey interferometry. Twice the time constant of this decay is our $T_2^{\mathbf{ECHO}}$ because $2T$ is the total time of free evolution and dephasing in our experiment.

3 Improving the code of the measurement procedure

3.0.1 The original setup

The experiments in the laboratory are executed in Python through Jupyter notebooks that use custom libraries to interact with the measurement equipment via the py-visa API. A single computer controls the measurement setup and is connected to the experimental devices through a regular IP-addressed network. The data is recorded into npz files saved directly on this computer. The equipment used to refrigerate and pump the vacuum in the experiment's chamber is separate from this measurement setup.

3.0.2 Modifications

The procedure to improve the code begins with applying good coding practices in Python for data analysis, such as avoiding for loops when a library function is already available (e.g., using NumPy's fill method instead of assigning each element in an array). It also involves avoiding repetition and unnecessary code, which increase the likelihood of mistakes, reduce efficiency, and hinder modification.

Other modifications included using the Plotly library for plotting instead of Matplotlib, because Plotly graphs are interactive and allow users to easily obtain information about their plots, such as the coordinates of a specific data point or fit, without having to manipulate the code. A trade-off is that although Plotly is more powerful, it is less intuitive to use.

The notebooks for the experimental procedures were also modified to follow a certain standard where the experiment parameters are separated from the execution in different code cells, effectively encapsulating them. However, this was not done completely, to allow users the flexibility to easily modify the execution if needed. If the procedures become reliable enough to be standardized, it would be beneficial to encapsulate them following Object-Oriented Programming principles, with an API to simplify the code used to run a series of measurements.

A data object for the experiment notebook was created to share parameters across all the different measurements. For example, the qubit transition frequency can be set once as a parameter and used in both the Rabi and Ramsey measurements. This prevents the user from having to modify the same parameter in multiple code cells and reduces the risk of mistakes from forgetting to update one or more parameters.

The saved files now record more metadata from the experiments. An API to retrieve the sample temperature could also allow this data to be stored for each data point.

Pulse Libraries In the laboratory's code for controlling the switch, I created new types of pulses (inheriting from the Pulse class) to be used directly by the switch without a mixer: the `GaussianCosPulse` and the `GaussianBorderCosPulse`. Both contain the envelope and the oscillation needed to excite the qubits; the former has a cosine with a Gaussian envelope, while the latter has a rectangular envelope with half-Gaussian curves on the borders. Previously, other pulses were implemented by generating the curve directly based on an absolute time vector in the `build` method. I reorganized this by adding an `envelope` method that takes a relative time vector as input (starting at 0 up to the total pulse length as a function of t). This method is then called in the `build` method together with the absolute time oscillation. The absolute time for oscillation is necessary to maintain phase coherence between different pulses by giving them a common phase reference. Separating the envelope from the oscillation improves code readability and makes it easier to implement modified versions with different envelopes based on the current code.

Also, for a pulse to be usable in the measurement procedure, its envelope must have a total area directly proportional to the pulse length. This way, if we increase the pulse length by some value, we know that the system will evolve proportionally to this increase. Therefore, (below are GitHub links to the code used to create these pulses) we can use a rectangular envelope pulse with envelope area $A{\cdot}$length, where A is the amplitude (SquareSideBandPulse), a gaussian pulse with area $A{\cdot}\sigma{\cdot}\sqrt{2\pi}$ where the lenght is related to σ (GaussianCosPulse) or even a sine wave envelope with area $\frac{2A\cdot length}{\pi}$ for $\sin(\frac{\pi t}{length})$ as envelope (SineEnvCosPulse). We can also do pulses that are a combination of those like GaussianBorderCosPulse.

4 The qubit excitation Mixer Characterization

The mixer is an RF minicircuit that uses a nonlinearity to multiply two signals. I characterized the mixer model IQ0307LXP. It multiplies the signal at the LO port with the signal from the I port and sums this signal with the product of the LO port and Q port signals, which has a 90º phase shift. In the time domain, we can say that the signal at the LO port is enveloped by the signals at either the I or Q ports, and both are

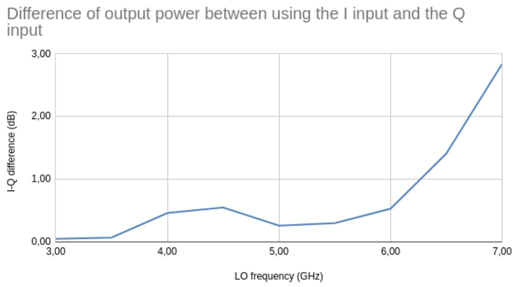

Figure 5: Difference in dB between the output obtained by using only the I input or only the Q input with the maximum AWG amplitude and 11 dB at the LO input. The graph displays the ratio (or subtraction in dB) of both values, $P_I - P_Q$.

summed. In the frequency domain, using trigonometric identities, we can verify that multiplying two cosines produces a signal containing cosines at the sum and difference of the two input frequencies (LO and I/Q), plus some harmonics and leakage of the original signals.

The signal in the LO port in our case is the main signal, a cosine at the qubit resonance frequency. We can then use pulses (usually square) in the I port to generate a zero-phase pulse of the qubit, pulses in the Q port to generate a 90º phase pulse, and any amplitude combination in I and Q to generate other phases. Nevertheless, we did not use this setup for the excitation of the qubit in my measurements; instead, we used the switch that was initially responsible only for generating the square pulses, due to leakage from the mixer interfering with the qubit state.

4.1 Leakage

We can verify in Figure 4 that the higher the frequency, the lower our leakage is. The LO amplitude does not make much difference.

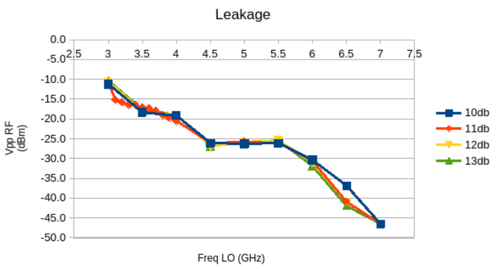

Figure 4: Leakage of the mixer using different LO signals with I and Q with no signal.

4.2 Signal/leakage ratio

We can diminish the influence of the leakage by attenuating the mixer output and using a high amplitude in the I and Q ports, which gives us a greater signal-to-leakage ratio. However, it is much more convenient to use the switch directly, since it does not have the leakage problem and can generate the necessary frequency. We also achieve better signal-to-leakage ratios at higher frequencies.

4.3 I and Q output amplitude difference

The higher the frequency, the greater the difference between the effects the two inputs, I and Q, have on the output. The user manual for the IQ0307LXP mixer advises using an attenuator to compensate for any difference that might exist on the input with higher output (for example, at 7 GHz we attenuate I by 3 dB, with Figure 5 as a reference).

5 The simulations

5.1 Measurement procedure with decoherence simulation

Here we reproduced the measurement procedures to characterize the qubit decoherence (Rabi oscillation, T_1 decay, Ramsey, and T_2 echo decay) in a simulation. The simulation was performed using QuTiP (Quantum Toolbox in Python) with the solver `mesolve` as its backbone. The parts with and without drive (such as the evolution of the Ramsey oscillation between the pulses) were separated to increase efficiency.

The notebook with all these simulations and detailed explanations is available here.

5.2 Drive white noise effect on decoherence

In this notebook, I adapted the code from the previous simulation to perform several tests on the impacts of background white noise in the qubit drive signal through simulations. I characterized the effect on the evolution of a driven qubit, on T_1, T_2^*, and T_2^{echo}. For the T_2 measurements, I separated the tests into noisy drive pulses and background noise during the evolution.

Then, I studied the effects on the system of white noise filtered by an ideal reject band filter around the qubit's resonance frequency, as well as a complementary band-pass filter.

5.3 Phase/Frequency Noisy Drive qubit simulation

In this notebook, we study the effects of noise in the phase of our drive signal. We input several different types of noise color (where color classifies a type of noise spectrum) into the phase of our signal. We also process these noises to simulate the behavior of a controlled wave generator. Additionally, we keep in mind the procedures of phase and frequency modulation from electrical engineering to understand the effects our disturbance has on the spectrum of the resulting signal.

5.4 Amplitude modulated low frequency noise drive qubit simulation

This notebook is more succinct because it is easier to observe the effects of low-frequency noises (brown and pink) modulated by the qubit resonance frequency as the carrier wave. This scenario could occur in a real experiment if there is some nonlinearity in the excitation circuit that mixes or multiplies the low-frequency noise with the drive signal. This does not matter for white noise since it covers the entire spectrum, and shifting it just generates a different white noise. However, for pink and brown noise or other low-frequency noises, if this process occurs, the majority of the noise will be near the qubit resonance frequency, where it has the greatest effect on the qubit.

Results and Discussion

6 Measurements

Given the constraints on time and the availability of the experimental equipment, I was unable to perform a systematic and thorough measurement of all samples. However, the combination covers most of the relevant properties of a qubit. For this final report, I removed some images to fit the page constraints, but you can see them in the partial report here.

6.1 The 2QTII sample

Consult the graphs notebook for more results and the vna measurement notebook and pulsed measurement notebook for the measurement code.

This was the first sample I measured; therefore, I followed the procedures more strictly, though without a full understanding of the function, goals, and expected results of each procedure. I also did not implement many improvements in the code at this stage. Those were gradually added as I identified opportunities for improvement using my Python skills gained in another internship. I was unable to perform the Ramsey oscillation here due to a change in the equipment setup—from using a wave generator source, then switching to a mixer and an AWG as the pulse generator, to directly using the AWG as the wave generator. The problem was that it was necessary to ensure the phases of each pulse in the Ramsey procedure were coherent.

Each sample was designed with certain values for its parameters, which will be listed before each sample section in the results. For this sample, the expected parameters calculated from the project's dimensions and parameters are 7 GHz for the cavity resonance and 4.38 GHz for the qubit resonance.

6.1.1 The punchout

From these measurements, we obtain the value of the cavity resonance at low power (dressed state) of 7.0207GHz, which will be used to calibrate the qubit measurement. We also chose an attenuation of 50 dB as the closest to the regime that best fits our approximations.

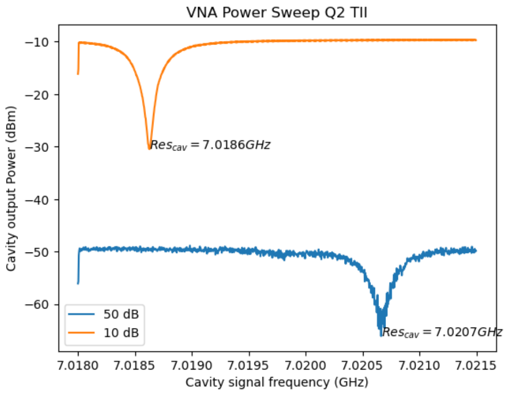

Figure 6: Frequency domain scan of the cavity for two different powers, with 50dB and 10dB attenuation, and respective resonance frequencies of 7.0207GHz and 7.0186 GHz. The 10dB case corresponds to high power and thus to the bare cavity resonance frequency, while the 50 dB case corresponds to the dressed cavity resonance.

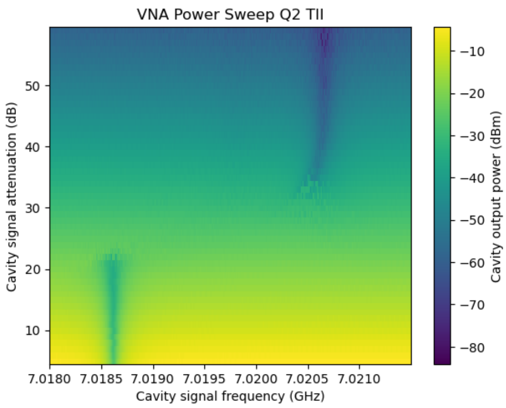

Figure 7: Frequency and power scan of the cavity in the frequency domain. The two vertical valleys correspond to the bare and dressed cavity resonance frequencies.

6.1.2 The Two Tone Spectroscopy

We got from this measurement the qubit resonance frequency for the first transition ω_{01} of 4.4540,GHz and for the two-photon transition from the ground to the second level $\omega_{02}/2$ of 4.1460,GHz, providing an anharmonicity to the qubit of $\omega_{01}-\omega_{12}=2\omega_{01}-\omega_{02}=$ 616MHz.

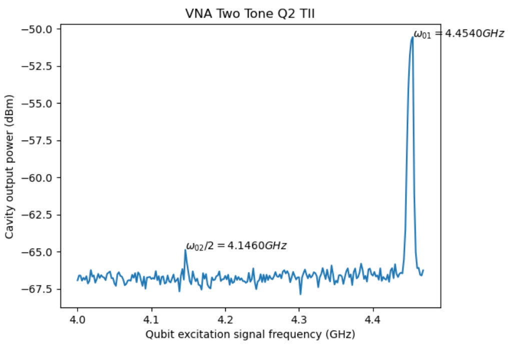

Figure 8: Frequency domain scan of the qubit excitation signal, with the y-axis representing the output amplitude of the cavity at cavity resonance. One can notice both the excitation to the first level of the qubit in the higher peak and the two-photon excitation from the ground to the second level in the lower peak.

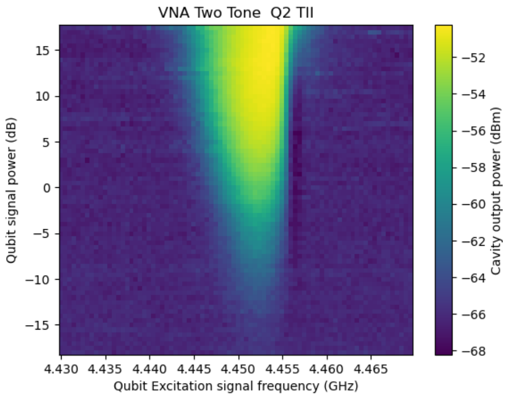

Figure 9: Frequency domain and power scan of the qubit excitation signal, with the color axis representing the output amplitude of the cavity at cavity resonance.

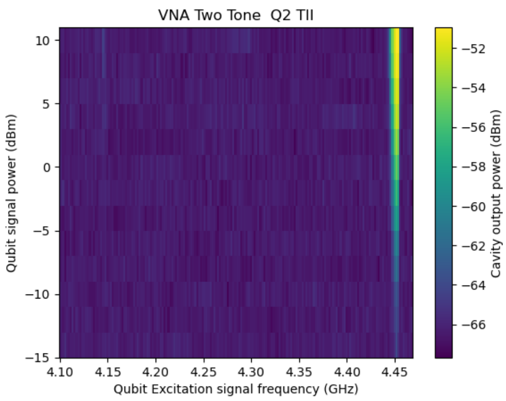

Figure 10: Frequency domain and power scan of the qubit excitation signal, with the color axis representing the output amplitude of the cavity at cavity resonance. One can notice, besides the main transition, the $0 \rightarrow 2$ transition, as seen in Figure 8.

6.1.3 The pulsed setup

In the pulsed setup, we obtain slightly different resonance frequencies: 7.0208GHz for the cavity and 4.4565GHz for the qubit. These values will be used for the remainder of the measurements in the pulsed setup.

6.1.4 The Rabi Oscillation

With the Rabi oscillation experiment, we find the length of the π pulse to be $\pi_{pulse} = 0.720\mu s$, which we can use to perform the other measurements.

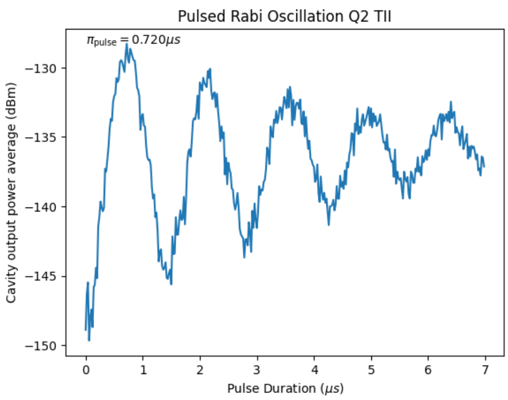

Figure 11: Rabi oscillation of the average power amplitude at the output of the cavity by varying the length of the qubit excitation pulse at its resonant frequency.

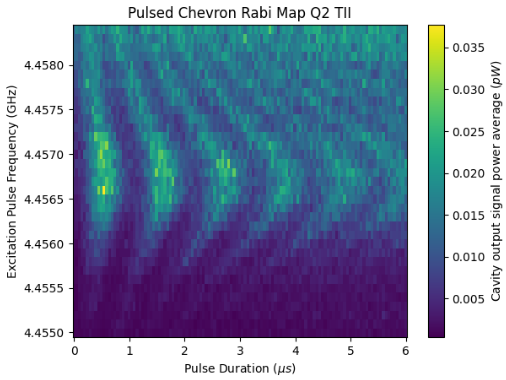

Figure 12: Rabi oscillation map of the average power amplitude at the output of the cavity (color axis) by varying the length of the qubit excitation pulse and the frequency corresponding to a detuning from the resonant frequency.

6.1.5 The T_1 measurement

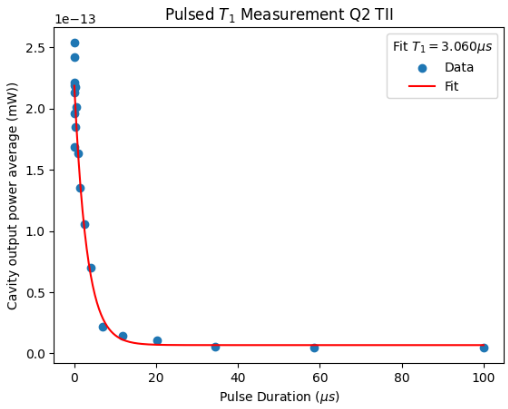

Figure 13: Measurement of the T_1 parameter of the qubit by waiting to measure a time T after a π pulse.

6.1.6 The T_2^{ECHO} measurement

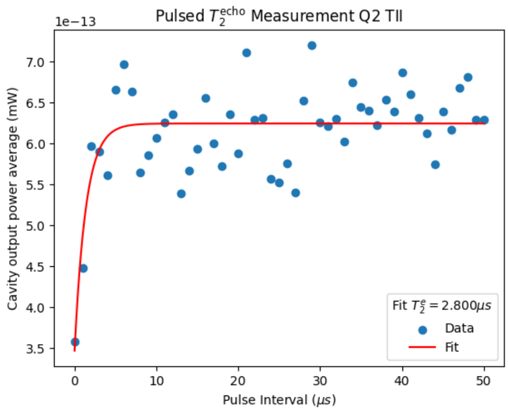

Figure 14: T_2^{echo} measurement made by applying a $\pi/2$ pulse followed by a π pulse and another $\pi/2$ pulse with intervals of length T between them.

6.2 The 3QTII sample

Consult the graphs notebook for more results and the pulsed measurement notebook for the measurement code.

This was the last sample I measured. I used the VNA data from measurements by another laboratory member to go directly to the pulsed setup measurements. Because I was more comfortable with the apparatus and the code, I was able to explore beyond the existing lab procedures. The problem with the Ramsey setup had already been solved, so I performed the Ramsey map, which can be seen below in the Chevron Ramsey section. In the

VNA punchout procedure, you can estimate χ from the difference in the cavity resonance. I used this estimation to measure the actual χ by finding the frequency to which the cavity shifts when the qubit is excited. With this shift, we have the experimental χ and can calculate the experimental value of the cavity-qubit coupling g.

The expected parameters for this qubit, calculated from the project's dimensions and parameters, are 7.1GHz for the cavity resonance and 5GHz for the qubit resonance.

I skipped the VNA part by using the one made by a laboratory colleague (Ednilson) on the same sample. He got the values: $f_{cavityatt=50dB}=$ 7.13954 GHz,$f_{cavityatt=10dB}=$ 7.14235 GHz, ω_{01} = 5.257 GHz, $\omega_{02}/2$ = 4.801GHz.

6.2.1 The pulsed setup

In the pulsed setup we get a slightly different resonance frequency for the 7.1421 GHz cavity and for the qubit of $\omega_{01}=5.2647GHz$. Those will be used for the rest of the measurements in the pulsed setup.

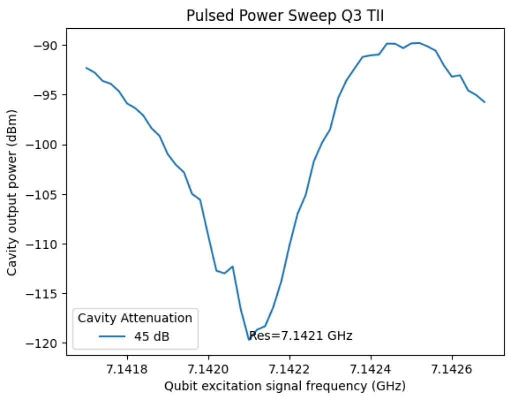

Figure 15: Frequency domain scan of the qubit excitation signal with the y-axis being the output amplitude of the cavity at cavity resonance for multiple attenuations in the cavity signal.

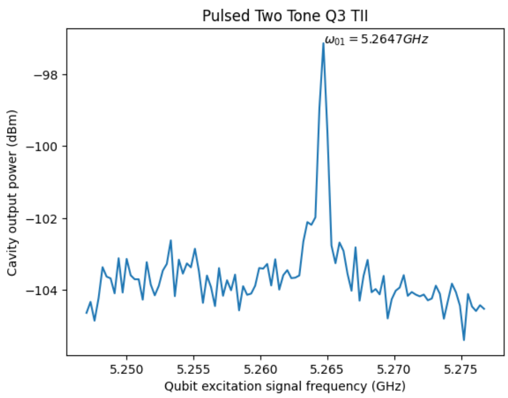

Figure 16: Frequency domain scan of the qubit excitation signal with the y-axis being the output amplitude of the cavity at cavity resonance.

6.2.2 The Rabi Oscillation

With the Rabi oscillation experiment, we find the length of the π pulse to be π_pulse=0.080,μs, which we can use for the other measurements. We aim for high power at the qubit to obtain a low Rabi period; however, power that is too high can heat the qubit stage of the dilution refrigerator.

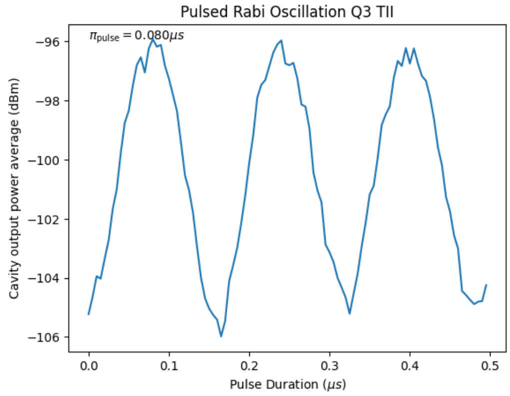

Figure 17: Rabi oscillation obtained by varying the length of the excitation pulse.

6.2.3 The T_1 measurement

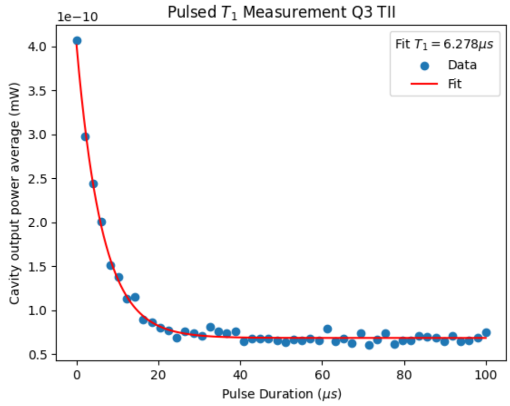

Figure 18: Measurement of the T_1 parameter of the qubit by waiting to measure after a pi pulse

6.2.4 The T_2^* measurement or Ramsey Interferometry

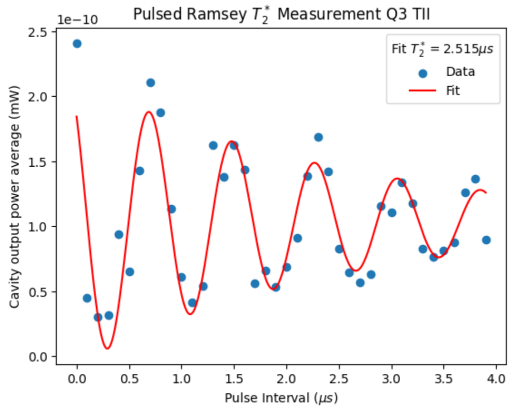

Figure 19: T_2^* measurement obtained by changing the distance between two pi/2 pulses.

6.2.5 The T_2^{ECHO} measurement

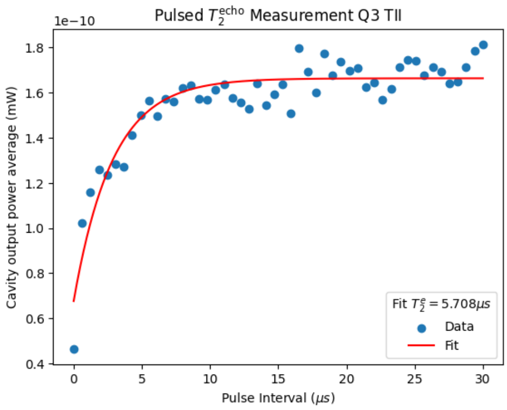

Figure 20: T_2^{echo} measurement made by applying a pi/2 pulse followed by a pi pulse and another pi/2 pulse with pulse intervals of length T between them.

Since the T_2^{echo} is typically larger than the actual T_2 and T_2^* is typically smaller we get the following approximate bounds for Q3's T_2 are $2.515\mu s \leq T_2 \leq 5.708\mu s$.

6.2.6 The position of the cavity resonance with an excited qubit

Here we find the position of the cavity resonance valley $f_{\text{cavity excited qubit}} = 7.14115\text{GHz}$, from which we can calculate the coupling parameters from the Jaynes-Cummings Hamiltonian in the dispersive approximation: $\chi = \frac{f_{\text{cavity}} - f_{\text{cavity excited qubit}}}{2} = 0.5250\text{MHz}$ and $g = \frac{\chi}{\Delta} = \frac{\chi}{f_{\text{cavity}} - f_{\text{qubit}}} = 31.40\text{MHz}$.

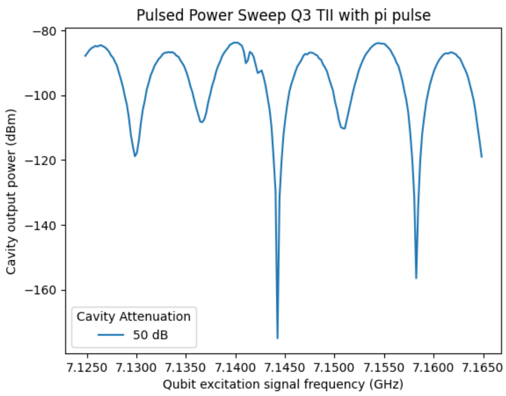

Figure 21: Cavity resonance valley with excited qubit with a broad range of frequencies.

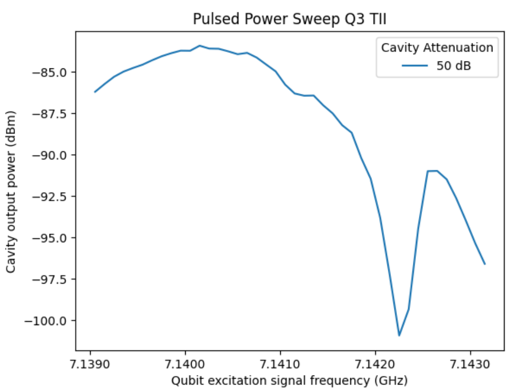

Figure 22: Position of excited qubit cavity resonance without excitation.

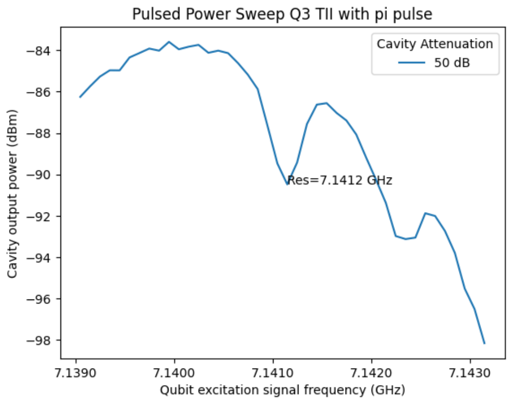

Figure 23: Cavity resonance valley with excited qubit. You can see that when the qubit is excited, contrary to 22 at the same range, it appears.

6.2.7 Chevron Ramsey

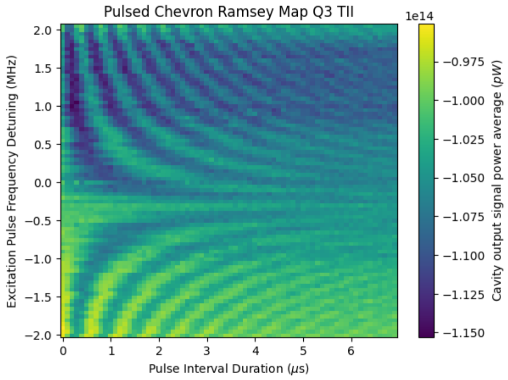

Figure 24: Ramsey map of the ramsey oscillation with varying detunings. The amount of detuning is equal to the oscillation frequency.

6.3 The 4QTII sample

Consult the graphs notebook for more results and the vna measurement notebook and pulsed measurement notebook for the measurement code.

This was my second sample. Although I had more experience and made some modifications to the code, there is not much difference in what I have done compared to the first sample, 2QTII. However, it was during this measurement that I discovered the cause of the problem with the Ramsey oscillation. I managed to perform the Ramsey oscillation, but the data does not look good, although the oscillation is visible and coherent with the theory. Since this particular sample required a lot of power in the qubit excitation pulse, my guess is that it also received more noise. In the noise simulations, we see that the Ramsey and echo procedures are much more sensitive to noise in the excitation pulses.

The expected parameters for this qubit, calculated from the project's dimensions and parameters, are 7.4GHz for the cavity resonance and 3.78GHz for the qubit resonance.

6.3.1 The punchout

From these measurements we get the value of the cavity resonance at low power (dressed state) of 7.443GHz that will be used to calibrate the measurement of the qubit. We also have chosen the attenuation of 50dB to be the closest to the regimen that best fits our approximations.

6.3.2 The Two Tone Spectroscopy

We got from this measurement the qubit's resonance frequency for the first transition ω_{01} of 3.963GHz and for the two photon transition from the ground to the second level $\omega_{02}/2$ of 3.808GHz. Giving an aharmonicity to the qubit of $\omega_{01} - \omega_{12} = 2\omega_{01} - \omega_{02} =$ 310MHz.

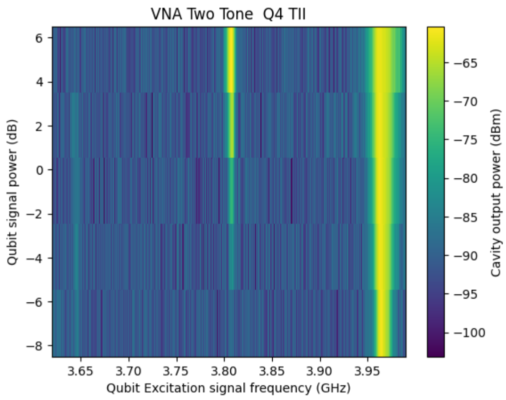

Figure 25: Frequency domain and power scan of the qubit excitation signal with the color-axis being the output amplitude of the cavity at cavity resonance.

6.3.3 The pulsed setup

In the pulsed setup we get a slightly different resonance frequency for the cavity of 7.444GHz and for the qubit of 3.9698GHz. Those will be used for the rest of the measurements in the pulsed setup.

6.3.4 The Rabi Oscillation

With the Rabi oscillation experiment we find the length of the π pulse to be $\pi_{pulse}=0.700\mu s$ which we can use to do the other measurements.

6.3.5 The T_2^* measurement or Ramsey Interferometry

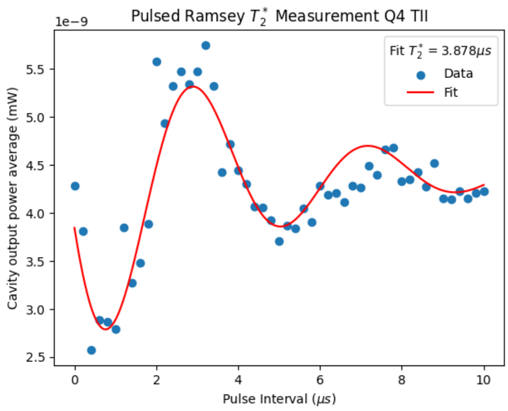

Figure 26: T_2^* measurement obtained by changing the distance between two pi/2 pulses.

7 Simulations

7.1 Measurement procedure with decoherence simulation

I successfully implemented a Rabi oscillation, Rabi map, T_1 measurement, Ramsey oscillation, Ramsey map and T_2^{echo} simulations using a lab frame Hamiltonian with an oscillating drive as a time dependent term in the total Hamiltonian. Using a rotating frame Hamiltonian would make it hard to ensure that the phase of two or more pulsed with free evolution intervals between them would have coherence phases. In the lab frame Hamiltonian it's just a matter of ensuring the sine waves used have consistent phase across all of the several steps of the simulation. This way also allows to reuse the code to for example model noise on the drive signal.

The simulation was also separated in steps when there were parts where the system freely evolved by getting the final state of a step and using it as the initial state of the next step. For example a step for the $\pi/2$ first pulse in the Ramsey oscillation, another step for the free evolution with length T and then a final step with the other pulse.

For the T_2 echo, it was found that numerical errors were causing additional unwanted dephasing. Decreasing the error tolerances of the 'mesolve' solver (atol and rtol options) improved this problem, as did using units that made the values in the simulation closer to one (e.g., 3,GHz instead of $3 \cdot 10^9$Hz) and normalizing frequencies, times, and energies with the qubit resonant frequency. All of this reduced

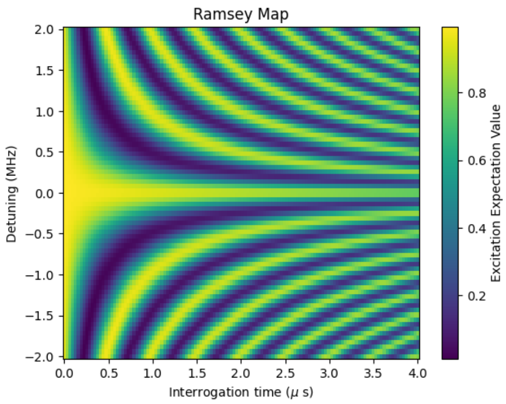

Figure 27: Simulated Ramsey map made in the measurement procedure simulation. Compare with figure 24.

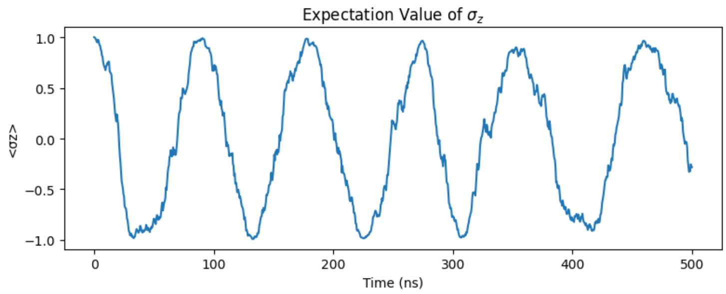

Figure 28: Rabi oscillation with 50ns *pi* pulse excitation drive and white noise with 10x as much standard deviation in amplitude of the drive amplitude

the dephasing caused by these numerical errors. This is elaborated in more detail in the Jupyter](https://github.com/Danielgb23 $simulation-qubit.ipynb$Jupyter) notebook.

7.2 Drive white noise effect on decoherence

The overall conclusion of this group of simulations is that the background noise in the drive does not significantly affect the process of relaxation (T_1). It causes the qubit to wander on the Bloch sphere, which will affect the process of measuring T_1, though the amplitude of the noise needs to be very high for it to be truly significant according to the simulation (several times the amplitude of an excitation pulse that can excite the qubit with a π pulse in 50 ns). The direction of this wandering is also in any direction, not unlike a random walk; the process of relaxation is necessarily towards the ground state of the qubit. The dephasing procedures, on the other hand, T_2, are much more sensitive to the background noise.

The idea that the part of the noise near the qubit resonance has the most effect while the qubit is robust against the rest of the spectrum can be confirmed by digitally removing a band around the resonance in the fast Fourier transform of the generated white noise, like an ideal reject band filter. Doing this in fact confirms this hypothesis through the simulations. We need noise with 100 times the 50 ns π pulse amplitude to have a significant effect. This effect is also different; instead of resembling a random walk in the σ_z expected value, it looks more like a white-noise-like high-frequency oscillation. Though in the real device, the Hamiltonian also couples to other frequencies: the qubit's higher energy transitions as well as the cavity harmonics.

7.3 Phase/Frequency Noisy Drive qubit simulation

The overall conclusion of these simulations is that the phase noise has more effect when it is low frequency (pink/brown noise). The low frequencies not only bring our signal out of phase during the excitation of the qubit but also off resonance when the noise signal is a slope of some sort. In the spectrum, it is the lower frequencies that get shifted near the qubit resonance when modulated, where they have the largest disturbance effect. The noise in the phase also redistributes energy away from the main signal to other points in the spectrum, increasing our Rabi period.

7.4 Amplitude modulated low frequency noise drive qubit simulation

If we understand the spectral effects of the noise on the qubit, this notebook works more as a confirmation of our understanding rather than adding something new. The pink noise, which has a more even distribution,

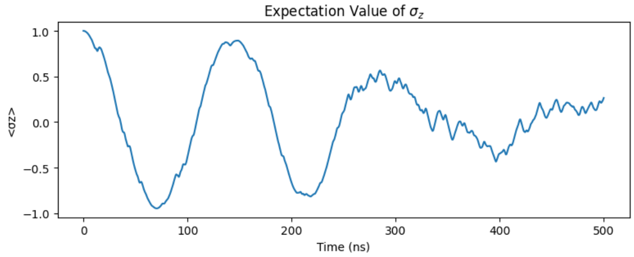

Figure 29: Rabi oscillation simulation with low pass filtered white noise in the phase of a resonant drive. Notice the noise driving the signal out of resonance in the end of the oscillation.

seems like a more potent white noise when modulated by a resonance carrier because it is concentrated at the resonance but not entirely. The brown noise is almost an excitation signal because it is all very low frequency, so it is very concentrated near the resonance when amplitude modulated.

We could, however, use band-filtered noise and modulate it with the resonance carrier to generate and study arbitrarily near-resonance noise.

Conclusion

During the first part of the internship, I developed a solid familiarity with the equipment and experimental procedures, alongside a clear understanding of the collected data and its interpretation based on quantum mechanics. Additionally, I gained insight into the limitations of purely quantum mechanical models in capturing real physical phenomena, particularly through the system's decoherence. The experiments allowed us to quantify this decoherence by extracting key parameters such as T_1 and T_2.

The second part, focused on simulations, initially served to reinforce and deepen the comprehension gained from the experiments. However, it soon expanded to leverage the student's background in electrical engineering within the computer engineering curriculum. By analyzing various noise types and manipulating their spectral characteristics, I was able to study their effects on the quantum system. This required integrating knowledge of signals, systems, and control engineering with the quantum mechanical concepts explored earlier.

This interdisciplinary integration, spanning quantum theory, experimental practice, simulations, and electrical engineering fundamentals was crucial for achieving a comprehensive understanding of qubit behavior under realistic conditions. It also underscored the value of combining diverse fields to address the practical challenges faced in quantum computing development.

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